

- **Objective**

The objective of this project is to analyze the Global Superstore dataset, identify important sales and profit patterns, build a regression model to predict sales, and create an interactive Power BI dashboard for business analysis and decision-making.

- **Dataset Description**

The dataset used in this project is the Global Superstore dataset.

It contains information related to orders, customers, products, sales, profit, quantity, discount, regions and categories.

The dataset was analyzed using Python and visualized using Power BI.

Q1 – EDA

Exploratory Data Analysis was performed using Python and Pandas.

The dataset was checked for:

- Dataset shape and columns
- Missing values
- Duplicate records
- Statistical summary
- Total Sales
- Total Profit
- Category-wise Sales

Visualizations were created to understand sales performance across different product categories.

Q2 – Regression Model

A Linear Regression model was created to predict Sales.

Features used:

- Quantity
- Discount
- Profit

Target variable:

- Sales

The dataset was divided into training and testing data.

The model was evaluated using MAE, MSE and R2 Score.

Model Evaluation Results:

MAE (Mean Absolute Error): 212.25

MSE (Mean Squared Error): 222185.03

R2 Score: 0.2730

Q3 – Power BI Dashboard

An interactive Power BI dashboard was created to analyze sales and profit performance.

The dashboard contains:

- Total Sales
- Total Profit
- Sales by Category
- Sales by Region
- Profit by Region
- Sales Trend Over Time
- Category Slicer

Q4 – Findings and Recommendations

Findings:

1. The total sales are approximately \$12.64 million.
2. The total profit is approximately \$1.65 million.
3. Among the product categories, Technology has the highest sales.
4. Western Europe has the highest sales among the regions shown in the dashboard.
5. Western Europe also shows the highest profit compared with the other regions.
6. The sales trend shows an overall increase from 2012 to 2015.

Recommendations:

1. The business should continue focusing on the Technology category because it generates high sales.
2. The successful sales strategies used in Western Europe can be analyzed for possible application in other regions.
3. Regions with lower sales and profit should be analyzed to identify areas for improvement.
4. The company should regularly monitor sales, profit, and discount levels using the Power BI dashboard.
5. The business should continue tracking the sales trend to support future planning and decision-making.

• Conclusion

This project helped analyze sales and profit performance using Python, Machine Learning and Power BI. Exploratory Data Analysis provided useful insights into the dataset, while the regression model was used to predict sales. The Power BI dashboard provides an interactive way to monitor sales, profit, category performance, regional performance and sales trends.